

Maximum Likelihood

Estimating parameters by optimizing probability

Open slides

Maximum likelihood estimates the parameter values that make the observed data most probable under a statistical model. The implementation pattern in this chapter is deliberately explicit: construct the likelihood, collect the parameters in one vector, and pass that vector to `optim()`.

Definition

For independent observations $y_1, \dots, y_n$ with density $p(y_i \mid \theta)$, the likelihood is

$$L(\theta; \mathbf{y}) = \prod_{i=1}^n p(y_i \mid \theta).$$

The maximum likelihood estimate is

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta; \mathbf{y}).$$

The product can become extremely small. Therefore we usually optimize the log-likelihood:

$$\log L(\theta; \mathbf{y}) = \sum_{i=1}^n \log p(y_i \mid \theta).$$

Because `optim()` minimizes, the function we pass to R is usually the negative log-likelihood:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \left[- \sum_{i=1}^n \log p(y_i \mid \theta) \right].$$

Normal Distribution

The normal density is

$$f(y) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{y - \mu}{\sigma} \right)^2 \right].$$

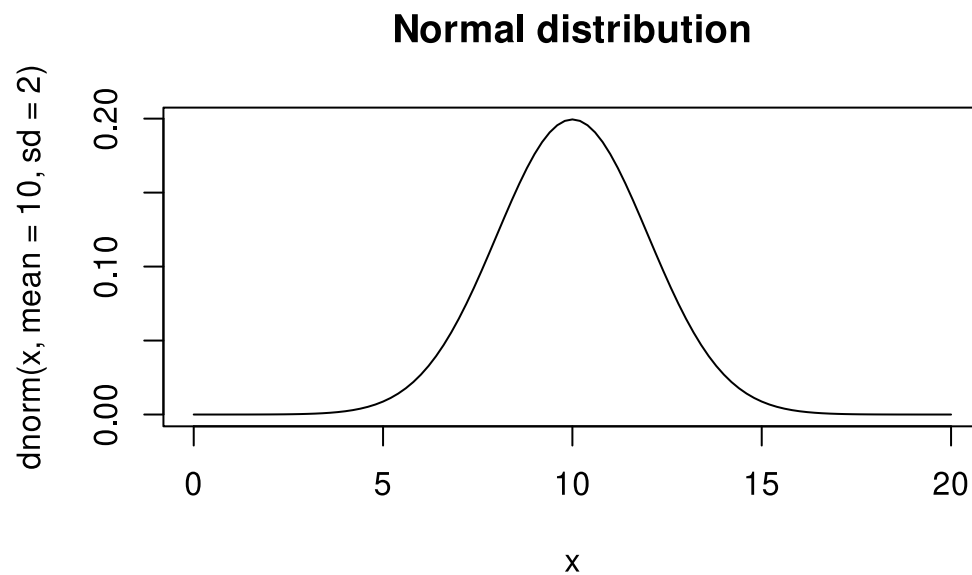
The normal distribution has two parameters: the mean μ and the standard deviation σ .

```
curve(  
  dnorm(  
    x,  
    mean = 10,
```

```

    sd = 2
  ),
  from = 0,
  to = 20,
  main = "Normal distribution"
)

```



Finding The Mean

Here σ is fixed at 1 and only μ is estimated.

```

set.seed(2026)

# generate random normal variable
mean = 4.5
sd = 1
n = 1000

y = rnorm(n, mean, sd)

# the likelihood function = L(y;.) = y ~ N(mu, sigma)
lik <- function(par, y, sd = 1) {

  # the parameter we want to estimate

```

```

mean = par[1]

# the likelihood of y
L <- dnorm(x = y, mean = mean, sd = sd, log = TRUE)

# the joint likelihood
return(-sum(L))

}

# the parameters we wish to estimate
p <- c(0)
names(p) <- c("mu")

# Get the Maximum Likelihood estimates for our unknowns
ML <- optim(p, lik, y = y)

# compare ML results to empirical mean
print("Maximum Likelihood:")

```

```
[1] "Maximum Likelihood:"
```

```
ML$par
```

```

      mu
4.514063

```

```
print("Empirical:")
```

```
[1] "Empirical:"
```

```
mean(y)
```

```
[1] 4.513767
```

Finding Mean And Standard Deviation

Now estimate μ and σ together. The only extra point is that σ must be positive, so the likelihood returns `Inf` for invalid values.

```

# likelihood for a positive standard deviation
sdlik <- function(x) {

  res <- 0
  if (x > 0) res = 1

  return(log(res))

}

# the likelihood function = L(y;.) = y ~ N(mu, sigma) * sigma ~ U(0, Inf)
lik <- function(par, y) {

  # the parameters we want to estimate
  mean = par[1]
  sd = par[2]

  # the likelihood of the standard deviation
  L2 <- sdlik(sd)

  # if sd is not allowed, stop here
  if (!is.finite(L2)) return(Inf)

  # the likelihood of y
  L1 <- dnorm(x = y, mean = mean, sd = sd, log = TRUE)

  # the joint likelihood
  return(-sum(L1 + L2))

}

# the parameters we wish to estimate
p <- c(0, 1)
names(p) <- c("mu", "sigma")

# Get the Maximum Likelihood estimates for our unknowns
ML <- optim(p, lik, y = y, method = "Nelder-Mead")

# compare ML results to empiricals
out <- rbind(
  ML = ML$par,
  empirical = c(mean(y), sd(y))
)

```

```
)
```

```
out
```

```
              mu      sigma
ML           4.513704 0.9893471
empirical 4.513767 0.9901296
```

Estimating Correlation

Now the likelihood is multivariate normal. The parameters are two means, two variances, and the correlation.

```
library(mvtnorm)

# likelihood on correlation coefficient.
# this is essentially a uniform prior on
# the range -1 to 1 and 0 density outside
# that range
corlik <- function(x) {

  res <- 0
  if (x >= -1 & x <= 1) res = 1

  return(log(res))

}

# the likelihood for the variance components.
# uniform above 0 and zero if negative
varlik <- function(x) {

  res <- 0
  if (x > 0) res = 1

  return(log(res))

}

# the likelihood function = L(y;.) = y ~ MVN(mu, sigma) * rho ~ U(-1,1)
lik <- function(par, y) {

  # the prior = likelihood of rho (correlation coefficient)
```

```

L2 <- corlik(par[5])

# and the priors on the variance components
L3 <- varlik(par[3])
L4 <- varlik(par[4])

# if any parameter is outside the allowed range, stop here
if (!all(is.finite(c(L2, L3, L4)))) return(Inf)

# we construct the covariance matrix out of the correlation coefficient
# and the variance components
covAB <- par[5] * (sqrt(par[3]) * sqrt(par[4]))
sigma <- array(c(par[3], covAB, covAB, par[4]), dim = c(2, 2))

# the likelihood of y
L1 <- tryCatch(
  dmvnorm(x = y, mean = c(par[1], par[2]), sigma = sigma, log = TRUE),
  error = function(e) -Inf
)

if (!all(is.finite(L1))) return(Inf)

# the joint likelihood
return(-sum(L1 + L2 + L3 + L4))
}

# those are the optimizers available
methods = c("Nelder-Mead", "BFGS", "CG", "L-BFGS-B", "SANN", "Brent")

# the parameters we wish to estimate
p <- c(0, 0, 1, 1, 0.2)
names(p) <- c("mu1", "mu2", "tauA", "tauB", "rho")

# create random data
set.seed(2026)
n = 10
tauA <- 3
tauB <- 10
rho = 0.7
covAB <- rho * (sqrt(tauA) * sqrt(tauB))
muA <- 5

```

```

muB <- 20

# the variance-covariance matrix
sigma <- array(c(tauA, covAB, covAB, tauB), dim = c(2, 2))

# the random data
Y <- rmvnorm(n, c(muA, muB), sigma)

# Get the Maximum Likelihood estimates for our unknowns
ML <- optim(p, lik, y = Y, method = methods[1])

res <- as.list(ML$par)
res$covAB <- res$rho * (sqrt(res$tauA) * sqrt(res$tauB))

# compare ML estimates with empiricals
out <- rbind(
  unlist(res),
  c(colMeans(Y), var(Y[, 1]), var(Y[, 2]), cor(Y)[1, 2], cov(Y)[1, 2])
)
colnames(out) <- names(res)
rownames(out) <- c("ML", "empirical")

# inspect results
out

```

	mu1	mu2	tauA	tauB	rho	covAB
ML	-22.484565	7.335499	70.379671	14.028532	0.8906419	27.985487
empirical	4.069481	18.482726	1.771725	9.252332	0.5003696	2.025885

Animal Model Likelihood

The animal model is

$$\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{a} + \mathbf{e},$$

with the marginal distribution

$$\mathbf{y} \sim MVN(\mathbf{X}\mathbf{b}, \mathbf{A}\sigma_a^2 + \mathbf{I}\sigma_e^2).$$

The implementation below keeps the construction explicit. We build $\mathbf{V} = \mathbf{A}\sigma_a^2 + \mathbf{I}\sigma_e^2$, calculate the multivariate normal log-likelihood, and minimize the negative log-likelihood.

```

library(mvtnorm)

```

```

set.seed(2026)

# the likelihood for the variance components.
# uniform above 0 and zero if negative
varlik <- function(x) {

  res <- 0
  if (x > 0) res = 1

  return(log(res))

}

# a small relationship matrix for a teaching example
n_animals <- 20
A <- diag(n_animals)
A[1, 2] <- A[2, 1] <- 0.5
A[3, 4] <- A[4, 3] <- 0.5
A[5, 6] <- A[6, 5] <- 0.25

# design matrix for the intercept
X <- matrix(1, n_animals, 1)

# true values used to simulate the data
b <- 30
var_a <- 4
var_e <- 9

# phenotype
a <- as.numeric(rmvnorm(1, mean = rep(0, n_animals), sigma = A * var_a))
y <- as.numeric(X %*% b + a + rnorm(n_animals, sd = sqrt(var_e)))

# likelihood function = L(y;.) = prod(PDF(y, E(y) = Xb, var(y) = A var_a +
I var_e))
lik <- function(par, y, X, A) {

  # likelihood for the variance components
  L2 <- varlik(par[1])
  L3 <- varlik(par[2])

  # if any variance is outside the allowed range, stop here
  if (!all(is.finite(c(L2, L3)))) return(Inf)

```

```

# it has to be this awkward because 'optim' only wants one vector of parameters
b <- matrix(c(par[3:(2 + ncol(X))]), ncol = 1)

# this is A var_a
G <- A * par[2]

# this is I var_e
R <- diag(length(y)) * par[1]

# this is var(y)
V <- G + R

# E(y)
mu <- X %*% b

# optim minimizes - therefore we need the negative likelihood
# Instead of taking the products over the likelihoods we take
# the sum over log(L) for numerical stability
L1 <- tryCatch(
  dmvnorm(x = y, mean = as.numeric(mu), sigma = V, log = TRUE),
  error = function(e) -Inf
)

if (!all(is.finite(L1))) return(Inf)

return(-sum(L1 + L2 + L3))
}

# This is only necessary because 'optim' wants one parameter vector
getp <- function(X, b = 0.1, var_e = 1, var_a = 1) {

  p <- c(var_e, var_a, rep(b, ncol(X)))
  names(p) <- c("var_e", "var_a", paste("b", 0:(ncol(X) - 1), sep = ""))

  p
}

# Get the Maximum Likelihood estimates for our unknowns
methods = c("Nelder-Mead", "BFGS", "CG", "L-BFGS-B", "SANN", "Brent")

```

```
ML <- optim(getp(X, b = mean(y)), lik, X = X, A = A, y = y, method = methods[1])

# compare ML results to the true simulation values
out <- rbind(
  ML = ML$par,
  true = c(var_e, var_a, b0)
)

out
```

	var_e	var_a	b0
ML	8.290905	7.775783e-06	30.09605
true	9.000000	4.000000e+00	30.00000

Exercises

1. Change the starting values in `p` for the normal mean example. Does the optimum change?
2. Estimate μ and σ for a smaller sample size, for example $n = 20$.
3. Change the true value of `rho` in the correlation example and rerun the analysis.
4. In the animal model example, change the relationship matrix `A`. What happens to the variance component estimates?
5. Explain why each likelihood function returns `-sum(L)` rather than `sum(L)`.